ECON 381: Midterm 1 — Answers

Professor Muyang Ren

Fall 2026

1. Multiple Choice Answers (30 points; 3 each)

Question	1	2	3	4	5	6	7	8	9	10
Answer	b	a	d	c	c	a	d	b	c	a

1. **(b)** The LLN states that $\bar{Y} \xrightarrow{p} \mu_Y$ under i.i.d. sampling with a finite mean.
2. **(a)** The fitted change is $(-2.28)(-2) = 4.56$ points.
3. **(d)** The CLT gives a standard normal limiting distribution for the standardized sampling error, not for individual observations.
4. **(c)** The population error collects other influences on test scores not represented by STR_i.
5. **(c)** The 99% interval uses the 99.5th standard normal percentile, approximately 2.58, leaving 0.5% in each tail.
6. **(a)** The conditional expectation is the population average within Los Angeles County.
7. **(d)** The sampling standard deviation of the mean is $\sigma_Y/\sqrt{n}$. Its estimated standard error and the resulting interval width generally decrease as sample size grows.
8. **(b)** OLS minimizes the sum of squared residuals.
9. **(c)** $\text{Corr}(X, Y) = \text{Cov}(X, Y)/(SD(X)SD(Y))$. Covariance magnitude depends on units and does not by itself determine correlation magnitude.
10. **(a)** The SER reports the typical residual size in the units of Y.

2. True or False Answers (20 points; 5 each)

(a) **False.** The population mean is fixed; the interval endpoints vary across samples. When the normal approximation is accurate, approximately 95% of the resulting intervals cover the fixed mean.

(b) **True.** s_Y measures dispersion among observations in the sample; $SE(\bar{Y}) = s_Y/\sqrt{n}$ estimates the sampling standard deviation of the sample average estimator under i.i.d. sampling.

(c) **True.** A correlation of 1 implies a perfect positive linear relationship (assuming finite, positive variances), so observations lie on an upward-sloping straight line.

(d) **False.** Multiplying every Y_i by 2, with X_i unchanged, doubles the OLS fitted values and residuals. Therefore, the residual sum of squares becomes $4RSS$, and

$$SER_{\text{new}} = \sqrt{\frac{4RSS}{n-2}} = 2SER.$$

The sample mean also doubles, so each deviation from the mean doubles and the total sum of squares becomes $4TSS$. Thus, provided $TSS > 0$,

$$R^2_{\text{new}} = 1 - \frac{4RSS}{4TSS} = 1 - \frac{RSS}{TSS} = R^2.$$

SER doubles, but R^2 remains unchanged. SER is measured in the units of Y, whereas R^2 is a unit-free measure of fit.

3. Analytical Answers (50 points)

The sample is $(X_i, Y_i) = (2, 16), (3, 10), (4, 10)$, with $n = 3$.

(a) **Sample means (4 points; 2 each).**

$$\boxed{\bar{X} = \frac{2+3+4}{3} = 3}, \qquad \boxed{\bar{Y} = \frac{16+10+10}{3} = 12}.$$

(b) **Sample standard deviation (5 points).**

$$s_Y^2 = \frac{(16-12)^2 + (10-12)^2 + (10-12)^2}{3-1} = \frac{16+4+4}{2} = 12, \qquad \boxed{s_Y = 2\sqrt{3}}.$$

(c) **Standard error of the mean (4 points).**

$$\boxed{SE(\bar{Y}) = \frac{s_Y}{\sqrt{n}} = \frac{2\sqrt{3}}{\sqrt{3}} = 2}.$$

(d) **Confidence intervals (8 points; 4 each).** Using $\bar{Y} \pm c\, SE(\bar{Y})$ and the critical values 1.65 and 1.96:

$$CI_{90\%} = 12 \pm 1.65(2) = \boxed{[8.70,\ 15.30]},$$
$$CI_{95\%} = 12 \pm 1.96(2) = \boxed{[8.08,\ 15.92]}.$$

Rounding note: Rounding the handout's displayed 1.645 gives 1.65. Rounding the unrounded standard normal percentile (approximately 1.64485) directly gives 1.64, yielding $[8.72, 15.28]$. Accept either 90% interval when the corresponding critical value is used consistently.

(e) **Coverage (4 points). No.** With only three observations, the CLT-based normal approximation may be poor. The procedure is not guaranteed to cover the population mean in 95% of repeated samples. The nominal confidence level describes repeated-sampling coverage when this approximation is accurate.

Analytical Answers (continued)

(f) **OLS slope (10 points).**

$$X_i - \bar{X} = (-1, 0, 1), \qquad Y_i - \bar{Y} = (4, -2, -2).$$

Therefore,

$$\hat{\beta}_1 = \frac{\sum_i (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_i (X_i - \bar{X})^2} = \frac{(-1)(4) + (0)(-2) + (1)(-2)}{(-1)^2 + 0^2 + 1^2} = \boxed{-3}.$$

The numerator is -6 and the denominator is 2.

(g) **Intercept and fitted equation (6 points).**

$$\boxed{\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X} = 12 - (-3)(3) = 21}, \qquad \boxed{\hat{Y}_i = 21 - 3X_i}.$$

(h) **Fitted values and residuals (6 points; 1 per entry).** Use $\hat{Y}_i = 21 - 3X_i$ and $\hat{u}_i = Y_i - \hat{Y}_i$:

i	X_i	Y_i	$\hat{Y}_i$	$\hat{u}_i$
1	2	16	15	1
2	3	10	12	-2
3	4	10	9	1

As a check, $\sum_i \hat{u}_i = 1 - 2 + 1 = 0$.

(i) R^2 **(3 points).**

$$TSS = (16 - 12)^2 + (10 - 12)^2 + (10 - 12)^2 = 24,$$
$$RSS = 1^2 + (-2)^2 + 1^2 = 6,$$
$$R^2 = 1 - \frac{RSS}{TSS} = 1 - \frac{6}{24} = \boxed{0.75}.$$

The line captures 75% of the total sample variation in Y.

4. Bonus Answers (5 points)

Here $(X_i, Y_i) = (3, 12), (2, 12), (4, 12)$, so $\bar{X} = 3$ and $\bar{Y} = 12$.

(a) **Slope with constant Y and comparison with constant X (3 points). Yes. The slope is well-defined and equals zero.** The centered regressor values are $(0, -1, 1)$, while all centered outcomes are zero. Therefore,

$$\hat{\beta}_1 = \frac{(0)(0) + (-1)(0) + (1)(0)}{0^2 + (-1)^2 + 1^2} = \frac{0}{2} = \boxed{0}.$$

The denominator is positive because X varies, so the zero numerator gives a unique, well-defined slope of zero. The intercept is $\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X} = 12$, and all fitted values equal 12.

Yes, the answer would change if X were constant and Y varied. With $X_1 = X_2 = X_3 = 3$, every $X_i - \bar{X}$ is zero. Both the numerator and denominator of the slope formula are zero, giving $0/0$ rather than $0/2$. The slope would then be undefined by the usual formula, and the data would not determine a unique slope and intercept.

(b) **R^2 with constant Y (2 points). No. The usual R^2 is undefined.** All outcomes equal their sample mean, so there is no variation in Y and $TSS = 0$. The fitted line matches every outcome, so all residuals are zero and $RSS = 0$. The numerator RSS and denominator TSS in the ratio below are both zero. Thus

$$R^2 = 1 - \frac{RSS}{TSS} = 1 - \frac{0}{0}$$

is undefined. Perfectly fitting a *constant* outcome does not give a well-defined R^2 under this formula, which requires $TSS > 0$.